Educational Quality – A Systems Perspective

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Quality means different things to different people. This presentation profiles a number of different systems perspectives on the concept of educational quality. Each perspective specifies different sets of criteria for assessing what organizational and/or educational quality means. Perceptions of, and decisions about, educational quality thus are context specific [15] --- there is no generalized 'recipe' for achieving educational systems quality (one size does not fit all). Responsibility for educational quality performance consequently should be left to individual schools and school districts.

Key Conclusions: (1) Seven different perspectives on assessing systems performance quality are addressed; (2) Five of these perspectives are directly relevant, and two indirectly relevant, to assessment of the quality of educational systems performance; (3) Quality performance criteria specified with these perspectives are categorized into input factors (influence on education and operational performance of educational systems) and output factors (longer-term effects beyond educational systems); (4) No two perspectives introduced here specify closely comparable sets of quality performance criteria; (5) There thus is no general 'magic bullet' for achieving educational quality; (6) Responsibility for quality performance should be placed primarily on individual schools and school districts.

1 Introduction

- What does the term 'quality' mean?
- We all know quality when we experience it, but describing and explaining it is a more difficult task [1] (defining pornography confronts the same challenge).
- Many people find quality an enigmatic concept. It is perplexing to define and often difficult to measure. One person's idea of quality often conflicts with another and, typically, experts often come to different conclusions when discussing what separates a higher from a lower quality school, college or university [1].
- Nevertheless, a number of perspectives on the relevance of quality to education and educational systems can be cited. A selective sample of these perspectives are reviewed, followed by some general conclusions.

2 Categorization of Educational Quality Perspectives

- Educational quality factors specified in the different perspectives addressed here are grouped into two categories.
- Input quality factors comprise those elements of educational quality that influence how well the educational system performs in educating its students, or in terms of operational effectiveness.
- Output quality factors influence longer-term outcomes beyond performance of the educational system itself --- i.e., quality outcomes for student careers and trajectories across the lifespan, for communities and societies, for local, regional, and national economies and governance, and/or for other modes of sociotechnical systems performance.
- The Thomas B. Fordham Institute perspective on educational quality [2] (Slde 12) similarly uses an input vs output categorization of educational quality factors, but its meanings of these two terms differ from the those above.

3 Following is a Summary of Various Perspectives on Input Educational Quality Factors Addressed Here

(These perspectives are not all linked to education, but they all are relevant to educational quality)

- Malcolm Baldridge National Quality Award in Education [3] (Slide 5)
- Educational ergonomic analysis [4-7] (Slide 6)
- ISO 9000 total quality management standard [1,8,9] (Slide 9)
- Educational systems ratings [10-12] (Slide 10)
- Thomas B. Fordham Institute criteria for educational quality [2] (Slide 12)
- Organizational systems rating criteria [14], 15 (Chap. 9)] (Slide 13)
- Principles of useful, effective instruction [13] (Slide 14)

4 Following is a Summary of Representative Perspectives on Output Educational Quality Factors

(note some overlap with input factors)

- Educational ergonomic analysis – early childhood education [4-7]
- ISO 9000 total quality management standard [1,8]
- Organizational systems rating criteria [14,15 (Chap. 9)]
- Investment in adult workforce education [16]

5 Malcolm Baldridge National Quality Award in Education [3]

- Most prestigious quality system in U.S.
- Since 2001, the Baldridge program has conferred nine educational quality awards at the elementary, secondary, and post-secondary levels.
- Criteria for a Baldridge National Quality Award in education focus on performance quality results of an educational system (i.e., input quality factors) in six areas:
  - student learning outcomes
  - customer satisfaction and engagement
  - product and service outcomes, and process efficiency
  - workforce satisfaction and engagement
  - budgetary, financial, and market results
  - social responsibility

6 Educational Ergonomic Analysis

Selected Ergonomic Design Factors Reliably Shown to Have a Strong Influence On Student Academic Achievement (i.e., input quality factors) [4-7]

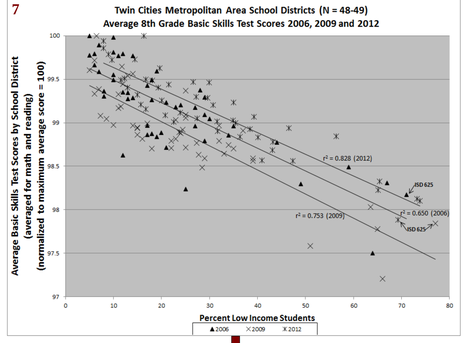
- Cooperative Learning
- Environmental Design of School Classrooms
- Longer School Days and/or School Year
- Later school day start times (for high school students)
- Physical Activity
- Teaching Quality
- Community Socioeconomic Status
- Early Childhood Education (also represents an output quality factor)

Arguably, a Quality Educational System is One that Addresses Most if not all of the Foregoing Factors

Among the Foregoing Factors, Community Socioeconomic Status has the Most Prominent and Robust Influence on Student Academic Achievement (see Slide 7 above)

Integrating Educational Ergonomics with Community Ergonomics Offers a Quality Approach for Mitigating the Adverse Impact of Low Socioeconomic Status on Student Academic Achievement [17,18]

7



8 Early Childhood Education (an output quality factor)

Few ergonomic design factors have been shown to exert a greater influence on the school performance (an input quality factor), and on the lifelong success generally (an output quality factor), of students than early childhood education [19].

9 ISO 9000 Total Quality Management Standard [1,8,9]

- Most widely adopted quality system, with over a million organizations certified to the system worldwide, and 25,101 certified systems in the U.S.
- Customers for this standard are production systems, not educational systems (I am unaware of any educational system certified to ISO 9000).
- Yet Sallis [1] points out that principles and practices of total quality management can be applied to educational systems.
- Certification criteria for the ISO 9000 standard with relevance to educational systems:

  - Management responsibility; - Statistical techniques; - Quality system (quality manual; quality procedures; quality planning); - Document & data control; - Quality records; - Training; - Contract review; - Design control; - Purchasing; - Process control; - Corrective & preventive action; - Internal quality audits

Relative to Non-certified Organizations, Organizations Certified to the ISO 9000 Standard Exhibit Higher Levels of Productivity and Competitiveness (input quality factors), and Longer Term Stability and Market Viability (output quality factors)

10 Educational Systems Rating Criteria [10-12]

- Various systems exist – my analysis will focus on two widely recognized approaches to assessing the input quality of educational systems --- the Common Core Standards and the Magnetic Schools of Excellence Standards
- The Common Core Standards (adopted by about three-fourths of U.S. states) comprise English Language Arts Standards, and Mathematics Standards
- The English Language Arts Standards focus on:
  - Reading; - Writing; - Speaking & Listening; - Language; - Media & Technology
- The Mathematics Standards focus on:
  - Mathematical Practice; and - Mathematical Content

11 Magnetic Schools of Excellence Standards

- Focus is on: - student academic achievement; - curriculum and teaching; - community engagement; - professional development; commitment to diversity

Premise

Compliance by an educational system (at some specified level) with the Common Core or Magnetic Schools of Excellence Standards suggests quality performance by the system

12 Thomas B. Fordham Institute (TBFI) Criteria for Educational Quality[2]

- The TBFI identifies five categories of educational quality: - Teacher quality; - Education input; - Education output; - Education social impact; - Education efficiency
- Within each of these categories, criteria for quality performance (at the State level) are specified (each based on state law and/or school policy)

Input quality factors

- Teacher quality (based on state law): teacher credentials, teacher responsibility for student achievement, hiring/firing law
- Education input: teacher salaries, pupil/teacher ratios, education cost per student
- Education output: % 4th graders at or above grade level in reading & math; mean ACT score for state
- Education efficiency: cost per student per % 4th graders above grade level in reading & math

Output quality factors

- Education social impact (statewide): per capita income; % population with college degrees; average library books checked out per capita
- median family income (statewide) positively related to all of the quality performance measures noted above

- Critique: Hassard [20] offers a biting criticism of the TBFI as nothing more than a mouthpiece for conservative propaganda concerning educational quality.

13 Organizational Systems Rating Criteria [14, 15 (Chap. 9)]

- Numerous rating systems have been developed to assess the operational effectiveness and quality of organizational systems
- I focus here on rating criteria advocated by van Reenen and colleagues at the London School of Economics, applied in a survey of 11,300 midsized firms in 34 countries. This by far is the largest sample size for any rating system I am aware of.
- This project graded the surveyed firms using a 5-point scale, based on three rating criteria, namely how the firms:
  - Monitored their operations
  - Set targets
  - Rewarded performance
- Key results --- a 1-point increment in the rating grade resulted in a:
  - 23% increase in productivity (input quality)
  - 14% increase in market capitalization (output quality)
  - 1.4 percentage points annual sales growth (output quality)
- This project did not survey educational systems. Arguably however, the rating criteria are relevant to educational systems quality.

14 Principles of Useful, Effective Instruction [13]

- The report specifies three sets of instructional design principles, which address the following three different functions of instruction:
  - memory (supported by six instructional design principles)
  - induction (support and guidance of learning (supported by 11 instructional design principles)
  - understanding (supported by 13 instructional design principles)

Premise

Instructors that Adopt these Principles in their Teaching Support Educational System Quality

15 Conclusions

- This analysis addresses 7 different perspectives on assessing systems quality (additional perspectives certainly exist). Five of these perspectives are directly relevant to assessment of the quality of educational systems performance (Slides 5-8, 10-12, 14). The remainder (Slides 9 & 13) arguably are indirectly relevant.
- Quality performance criteria specified with these perspectives are categorized into input and output quality factors. Input factors influence the educational and operational performance of the system. Output factors influence longer-term outcomes beyond performance of the educational system itself.
- No two perspectives introduced here specify identical --- or even closely similar --- sets of quality performance criteria. This relates to the point that different individuals have different views of quality.
- Thus, there is no general 'magic bullet' for achieving educational quality. Responsibility for quality performance should be placed primarily on individual schools and school districts.

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